Directed Energy Beam Propagation: Range-to-Effect Analysis

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3/15/2026

Foreward:

I have been an admirer and student of unmanned aerial systems, UAS, over the past few years in my time at university. That said I have grown equally interested in counter UAS systems, cUAS, as of late. However, the resources readily available for one to study such systems are greatly limited by comparison to UAS systems. This is especially true for directed energy systems. As a result I decided to do some research on my own initiative to begin building a foundational understanding of electronic warfare systems. I chose to focus on directed energy beam propagation with an emphasis on range to effect analysis. My work, findings, and results are as follows.

Abstract:

Directed energy systems make use of high energy laser beams to propagate through varying atmospheric conditions while maintaining power and intensity delivery to effect the target. However, the physical processes - ranging from diffraction to attenuation to thermal blooming, all limit the effective engagement range.

This project develops a Python based simulation that serves as a framework for modeling laser beam propagation and evaluating the range to performance effects under system and operating conditions. Using Gaussian beam optics, Beer-Lambert atmospheric attenuation, and an idealized thermal blooming simulation this model captures a range of beam to medium interactions

Several parametric trade off analyses were performed to evaluate the impact of various system parameters ranging from beam waist, to laser power, to atmospheric conditions. The results demonstrate the existence of an optimal beam waist for a given atmospheric environment and highlight diminishing performance returns as laser power is increased.

Introduction and Objectives:

Directed energy systems are increasingly being explored for a range of applications including but not limited to: missile defense, cUAS, and long range precision engagement. These high energy laser systems offer unique asymmetric advantages allowing for defensive engagements at the speed of light, a virtually endless magazine, and high munitions efficiency. However, these systems in their current form are highly dependent on atmospheric propagation.

This limitation presents the greatest challenge in effectively deploying directed energy systems. Laser beams propagating through an atmospheric medium experience several degradation mechanisms. Diffraction causes the beam to diverge as it propagates, leading to a reduction in on-axis irradiance with increasing distance. Atmospheric absorption and scattering result in the reduction of transmitted power. In high intensity systems thermal blooming can occur if the beam waist is too tight or the atmosphere is especially conducive to thermal heating. This atmospheric heating leads to an increase in the refractive gradient that further distorts and degrades the beam.

My objective in this project is to create a computational model capable of demonstrating the analysis of system design parameters, like beam waist and laser power, and how they interact with atmospheric conditions to impact the effect range and abilities of the laser system.

Modeling Approach:

The propagation of high energy beams through an atmospheric medium is governed by several physical processes that influence the travel of the beam, the behavior through the medium, and the energy delivered onto the target. In the scope of this project a simplified first order model was developed to capture the effects limiting and controlling directed beam performance. This model uses Gaussian optics to describe beam characteristics and Beer-Lambert analysis to represent atmospheric conditions. The modeling workflow approach follows a sequential pathway in which the beam is analyzed along its path for propagation, atmospheric losses, and beam-medium interactions.

Gaussian Beam Propagation: Laser beams produced by high energy laser systems can be approximated as Gaussian beams. The transverse intensity distribution of a Gaussian beam decreases exponentially away from the beam axis.

The radial intensity profile of a Gaussian beam is given by:

$$I(r,z) = I_0(z)\, e^{-\frac{2r^2}{w(z)^2}}$$

Equation 1: Radial Intensity Distribution

Where:

I(r,z) = irradiance at radial position r and distance z

I_o(z) = peak on-axis irradiance

w(z) = beam radius at distance z

r = radial distance from beam center

Here beam radius evolves with the distance according to the Gaussian proportion law:

$$w(z) = w_0\sqrt{1 + \left(\frac{z}{z_R}\right)^2}$$

Equation 2: Beam Radius Evolution

Where:

w_0 = beam waist radius

z_R = Rayleigh range

z = propagation distance

The Rayleigh range designs the distance over which the beam radius increased by a factor of sqrt(2) and is given by:

$$z_R = \frac{\pi w_0^2}{\lambda}$$

Equation 3: Rayleigh Range

Where lambda is wavelength. At distances where the engagement range is larger than that of the Rayleigh range the beam expands approximately linearly with distance.

Beam Divergence: The far-field divergence angle of a diffraction limited Gaussian beam is approximated by:

$$\theta = \frac{\lambda}{\pi w_0}$$

Equation 4: Beam Divergence

Where theta is the far-field half-angle divergence. This strongly demonstrates the relationship between large beam waist and smaller divergence angles. This leads to the conclusion that you can increase the beam waist in order to improve long range beam quality by reducing diffraction spreading. It should be noted that by increasing the waist to reduce the initial beam intensity exemplifying the tradeoff between intensity and divergence.

Atmospheric Attenuation: As a beam propagates through an atmospheric medium it experiences power losses due to absorption and scatter by atmospheric constituents. In this model I have used Beer-Lambert Laws to represent these losses.

The power transmitted by the beam at some distance “z” is given by:

$$P(z) = P_0\, e^{-\alpha z}$$

Equation 5: Power After Propagation

Where:

P(z) = remaining beam power

P_o = initial beam power

alpha = atmospheric attenuation coefficient

z = propagation distance

The attenuation coefficient represents the combined effects of: molecular absorption, aerosol scattering, and atmospheric particulates. Higher attenuation coefficients correspond to degraded atmospheric conditions.

Beam Irradiance: The peak irradiance along the axis of travel is determined in large part by the transmitted power and beam cross sectional area, i.e. beam waist.

The peak irradiance is given by:

$$I_0(z) = \frac{2P(z)}{\pi w(z)^2}$$

Equation 6: Peak Irradiance

The average irradiance across the beam cross section is:

$$I_{\text{avg}}(z) = \frac{P(z)}{\pi w(z)^2}$$

Equation 7: Average Irradiance

These relations display that irradiance decreases as the beam radius increases even if transmitted power remains constant. As you spread out the energy delivered on target the irradiance must decrease.This is true for a change in beam geometry or as a result of atmospheric attenuation.

Thermal Blooming: At high intensity operation the beam can heat the medium along the propagation pathway. As a result a change in air density occurs in the space immediately local to the beam. What follows is a gradient in the refractive index of the medium. This refractive index behaves as a negative lens resulting in a beam that spreads out much more rapidly than one in a simple diffraction environment. In the model here this blooming effect is represented by a thermal blooming coefficient.

The effective beam radius is approximated by:

$$w_{\mathrm{eff}}(z) = \sqrt{w(z)^2 + w_{\mathrm{bloom}}(z)^2}$$

Equation 8: Effective Beam Radius

The blooming term is modeled by:

$$w_{\mathrm{bloom}}(z) = k_b \sqrt{P_{\mathrm{air}}}\, z$$

Equation 9: Blooming Term

Where:

k_b = blooming coefficient

P_air = laser power absorbed by the atmosphere

The blooming term used here is an empirical approximation intended to represent beam spreading caused by refractive index gradients induced by atmospheric heating. A full treatment would require solving coupled Navier–Stokes and heat transfer equations.

Target Energy Absorption: The energy delivered on target is dependent on the amount of power absorbed by the target surface over the period of time the beam is kept on target. This is materially driven as some materials absorb more energy than others. To model this I assigned a target absorptivity coefficient.

The absorbed power is modeled as:

$$P_{\mathrm{abs}} = A\, P(z)$$

Equation 10: Absorbed Power

Where A is the target absorption coefficient. The total energy at exposure time “t” is:

$$Q = P_{\text{abs}}\, t$$

Equation 11: Energy Delivered to Target

Target Temperature Rise: the temperature increase of the target is estimated using a simple single lumped thermal model. Within this model temperature rise is given by:

$$\Delta T = \frac{Q}{m c_p}$$

Equation 12: Target Temperature Rise

Where “m” is the heated target mass and “cp” is the specific heat capacity of said target mass. This simple approximation provides a direct first-order estimate of the thermal effect delivered onto the target surface.

Range-to-Effect Metric: The effectiveness of the directed energy system is evaluated using a range-to-effect criteria. Effectiveness is determined if the peak irradiance exceeds a threshold radius. Here the maximum effective engagement range is now defined as the largest distance where this condition can be satisfied. With this metric we now have a way to convert beam propagation physics into a common performance measure.

$$R_{\text{effect}} = \max\left\{ z \mid I_0(z) \geq I_{\text{threshold}} \right\}$$

Equation 13: Effective Range Determination Condition

Systems Parameter Table:

Parameter	**Symbol**	**Value**	**Notes**
Laser Wavelength	λ	1.06 μm	Typical fiber laser
Beam waist	w_0	5–50 mm	Sweep range
Laser power	P_0	1–20 kW	Sweep range
Attenuation coefficient	α	e.g. $10^{-4}\ \mathrm{m}^{-1}$	Clear air
Target absorptivity	A	0.6	Metal target
Specific heat	c_p	—	Target material

Simulation Workflow:

1.) Define beam parameters: P0,w0,λ
2.) Compute Rayleigh Range
3.) Step through propagation distance *z*
4.) Apply atmospheric attenuation
5.) Compute beam radius and irradiance
6.) Apply blooming correction
7.) Evaluate range-to-effect condition

Results and Analysis:

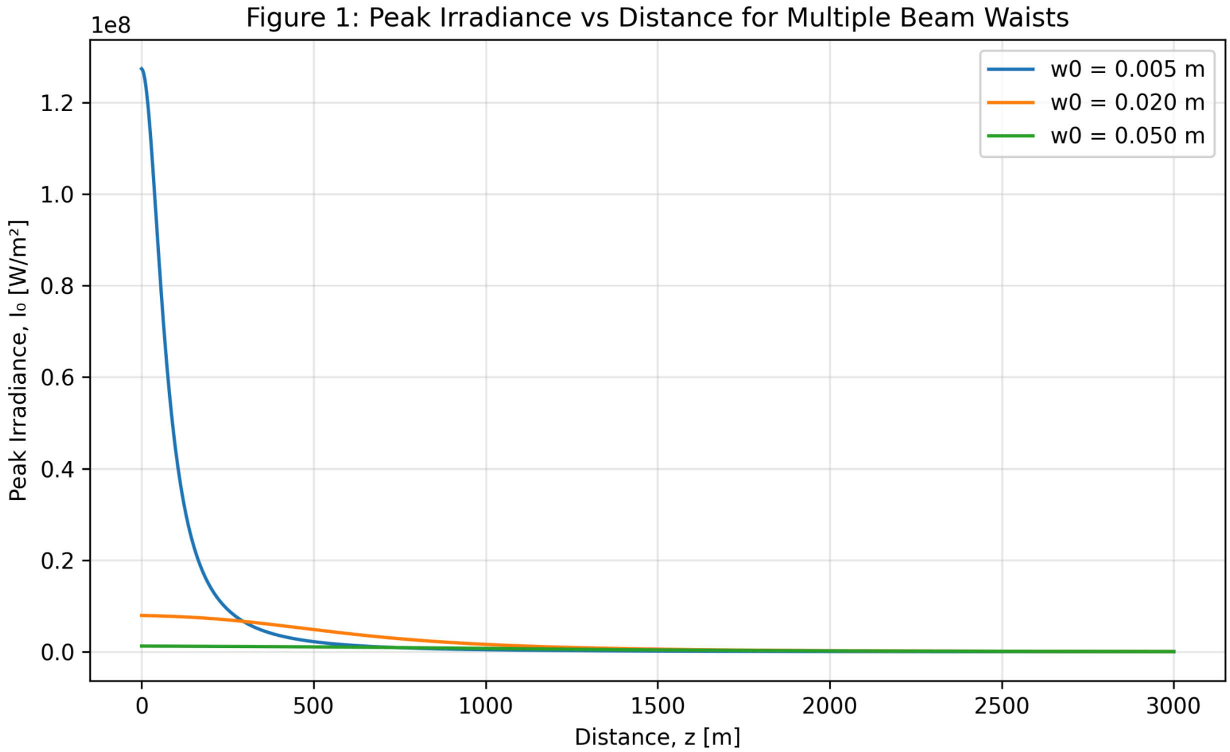

Figure 1

Peak Irradiance in W/m^2 vs. Distance in meters for a range of waist length values. Smaller beam waist produce a higher initial intensity while falling victim to divergence more rapidly, thus reducing long range performance.

This figure gives us key insights on laser waist and effective target range. A small waist allows for high intensity but results in rapid divergence and ineffectiveness at longer ranges. By comparison a larger waist allows for lower divergence behavior but simultaneously a much weaker intensity on target.

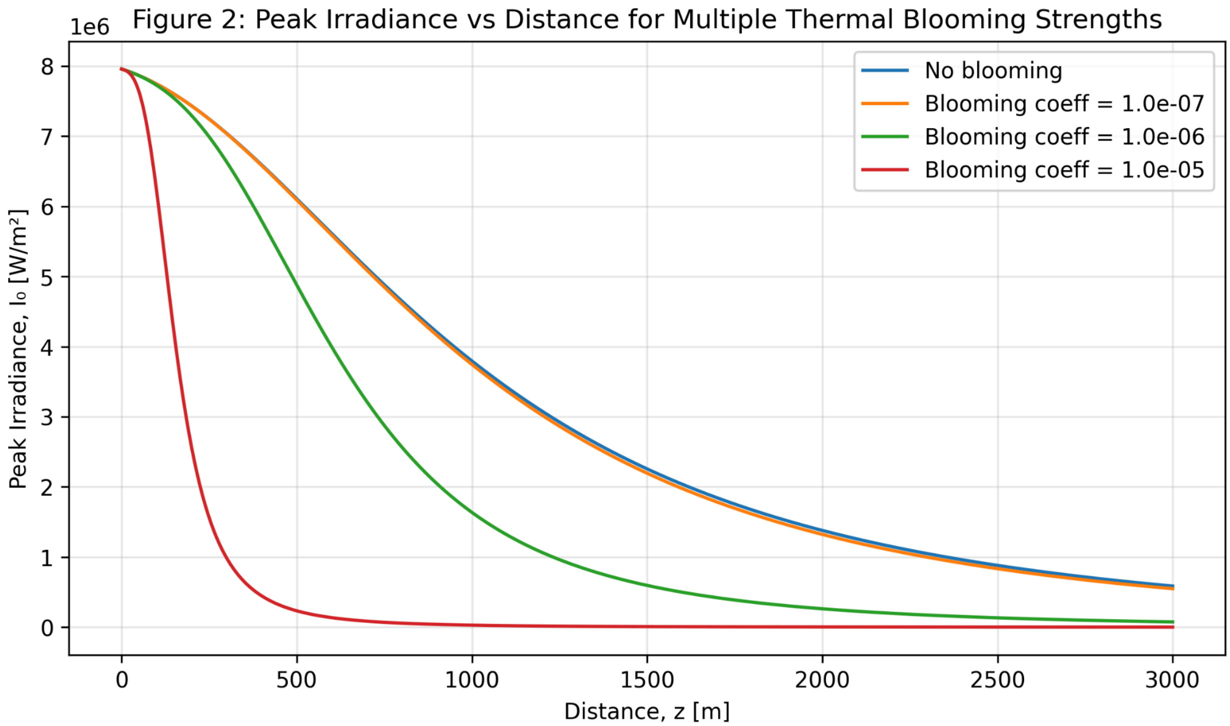

Figure 2

Impact of thermal blooming on peak irradiance during propagation.

As the blooming coefficient increases, and consequently atmospheric attenuation, the spreading of the beam increases as well. As a result, long range irradiance collapses at a faster rate. This also begins to demonstrate how beam-medium interactions limit on target performance.

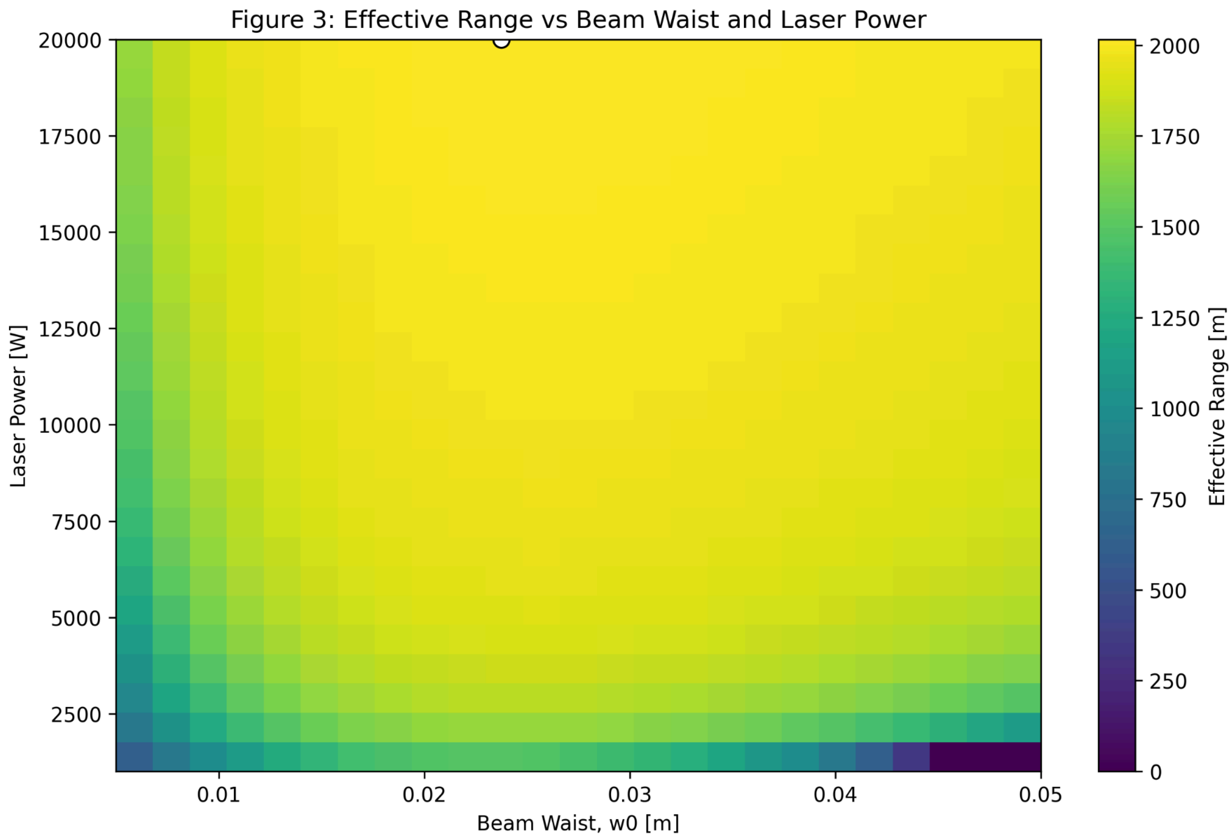

Figure 3

Effective engagement range as a function of beam waist and laser power.

This figure allows for the identification of an optimal waist region that allows beam waist and laser power to be optimized for an effective range. Key limitations are expressed as well, mainly low power limits and large waist effects. The diminishing returns of overly high power are also explored as the same effective range can be reached for much less energy expenditure.

Conclusions:

This project demonstrates important behaviors governing directed energy propagation. I have learned a strong foundation in Gaussian Optics alongside Beer-Lambert Atmospheric analysis. My key takeaways from this project have been:

- There is a beam waist that balances beam divergence and intensity with a given power.

- Increasing laser power improves engagement range but strongly exhibits diminishing returns due to atmospheric propagation effects.
- Thermal blooming significantly degrades beam quality in high intensity cases.

It is clear that the optimization of beam geometry and system power are critical to not only the design of the energy system but also the deployment and target engagement of the system as well. This said the model I have created does come with several key limitations. This model does not account for atmospheric turbulence, adaptive optics, or high level computation fluid dynamics. I have considered forming a follow-up study that integrates computational solutions for these limitations with a focus on adaptive optics and CFD.

Afterword:

I hope this write up and the accompanying notes and code scripts demonstrate a strong understanding of the foundation of Gaussian optics and Beer-Lambert atmospheric optimization. I was inspired to pursue this undertaking by Aurelius Systems and Eprius respectfully. Thank you for reading if you came this far. I hope you found my work to be sound and correct. Please feel free to contact me with corrections, comments, and anything else.

Thanks,

John C. Richards